Big Data to Knowledge(BD2K)

# Best Practices for Biomedical Research Data Management

## Module 5: Data Management Policy
## Data Policy

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This video will define the concept of policy as it relates to research data management. It gives an overview of the distinction among National Policy Directives, funder policies, institutional policies and publisher policies.

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# Policy

"A principle of behavior, conduct etc. thought to be desirable or necessary, especially as formally expressed by a government or other authoritative body."

*Wiktionary. "Policy" Available from: https://en.wiktionary.org/wiki/policy.*

"Typically instituted to avoid some negative effect…or to seek some positive benefit."

*Wikipedia. "Policy" Available from: https://en.wikipedia.org/wiki/Policy*

# Public Data are a Public Good

*OECD. Principles and Guidelines for Access to Research Data from Public Funding. Paris, France: Organization for Economic Cooperation and Development; 2007. Available from: http://www.oecd.org/sti/sci-tech/38500813.pdf.*

1. Openness
2. Flexibility
3. Transparency
4. Legal conformity
5. Protection of intellectual property
6. Formal responsibility
7. Professionalism
8. Interoperability
9. Quality
10. Security
11. Efficiency
12. Accountability
13. Sustainability

# Public Data are a Public Good

"Publicly funded research data are a public good, produced in the public interest, which should be made openly available with as few restrictions as possible in a timely and responsible manner that does not harm intellectual property."

RCUK Common Principles on Data Policy (2011)

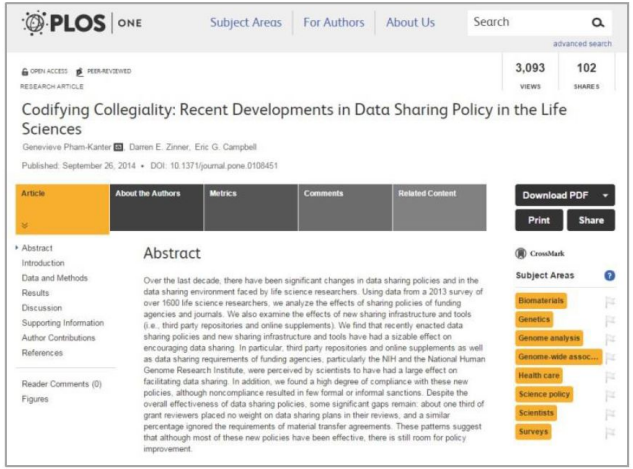
"The Administration is committed to ensuring that, to the greatest extent and with the fewest constraints possible…, the direct results of federally funded scientific research are made available to and useful for the public, industry, and the scientific community."

John Holdren, OSTP Memorandum for the Heads of Executive Departments and Agencies (2013)

Shared Principles
OECD
National Directives
Data sharing; reproducibility
Funder Policies
Data management
Institutional Infrastructure
UMASS
Verification; peer review
Investigator Practice
YOU

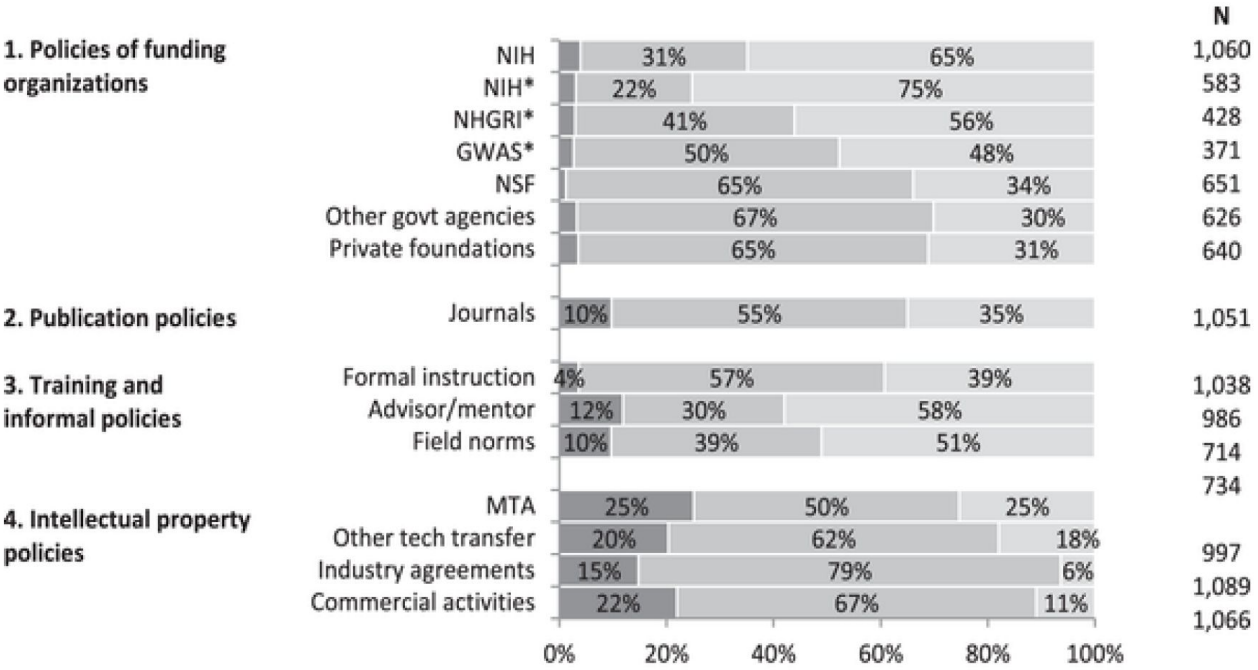
# Impact of Policies

- Survey of 3000 life scientists (41% response rate)
- Impact of data sharing policies in the life sciences since 2000
- Policy types include
  - Federal and Private funders
  - Journals
  - Institutions
  - Informal practice




*Pham-Kanter G, Zinner DE, Campbell EG (2014) Codifying Collegiality: Recent Developments in Data Sharing Policy in the Life Sciences. PLoS ONE 9(9): e108451. doi:10.1371/journal.pone.0108451*
http://www.plosone.org/article/info:doi/10.1371/journal.pone.0108451

# Figure 1. Influence of policies on data sharing.

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# Impact of Policies

Policies of the NIH have had the greatest impact on facilitating data sharing (65% of respondents).

- Effectiveness of policies may lie in reducing the administrative and procedural burdens of sharing. (IE: establishing dbGAP and other repositories; “leveling the playing field”)

- -1/3 reviewers did not consider data sharing plans important
- Few sanctions or penalties for non-compliance
- +1/3 violated or did not know institutional policy

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